Answer Part 0

Team and Evidence Index

StudyTrack - Lab5 Evidence Index | 13 November 2026

Team **TH-057** | Section **B** | LEB2 submission group **TH-057_P8**

Group name: **กำลังปั่นเครดิตครับ**

Topic **8: Habit and Activity Tracking System**. Manager activity templates, personal daily reading goals and saved minutes/progress, with private ownership boundaries.

Student	Student ID	GitHub	Role
Pitchai Chadchuangchot	67070501068	Datakung	B
Phanuwit Butchari	67070501070	auto4496	A

Public repository: auto4496/studytrack. **Public application:** open StudyTrack.

Application source selected for assessment: **6d30aefece55b3a83194010a2e0e62c9b8b1dbfc** (main release), observed serving via Deploy43410699104. Accepted evidence snapshot: **c6b63f733489b8390c9b278351944b67937f0d84**, after PR53/B acceptance. The corrected report, index and runbook were accepted in PR56.

Evidence index (direct checks)

Check / part	Primary direct evidence	Actual result
C-01 / Part1	PR52, CI43409691975	Meaningful CI SUCCESS:41 offline, 152 API/unit, 38 browser cases; app/both image builds.
C-02 / Part1	PR36, failed run42781653763	Peer approval present; required failure still blocked merge. Closed unmerged.
C-03 / Part2	Container manifest, PR15	Fresh clone/no-cache build,35 actual production/lifecycle checks; later guarded correction.
C-04 / Part3	P0, P2/privacy	Own save/read/refresh; user-operated second account cannot read private record.
C-05 / Part3	P1 same-image run, P2 comparison	Full original DTOs equal after process replacement and distinct visible redeploy.
C-06 / Part4	PR51 -> PR52 -> Deploy43410699104	Separate peer-reviewed staging/main path, actual pipeline and serving revision.

Complete per-requirement links and evidence limits are in the source evidence index. The following page maps R-01 through R-12 and actual contributions.

Answer Part 0 - traceability and contribution

Requirement	Evidence	Result / limit
R-01	C-03; clean build log	Historical exact source ee56d91, not relabelled as a new build of today's checkout.
R-02	C-03; settings table/runbook	Placeholders/private settings separated; runtime/tools/config isolation documented.
R-03	C-04; public URL, Figures3-4	Business save and owner-boundary denial on the live service.
R-04	C-05; Figures5-9, full DTO comparison	Same original IDs,stored date,count,timestamps,20/30 and66.67% at P0/P1/P2.
R-05	C-01; CI workflow	PR CI runs meaningful isolated unit/API/browser/build checks.
R-06	C-02; Figures1-2	Required failure blocked an approved normal PR; no bypass attempt.
R-07	C-06; deploy workflow	Reviewed main push built/migrated/deployed/smoked real releaseB.
R-08	C-06; procedure	Full source/tree/digest/run/revision mapping and repeatable steps.
R-09	Lab5 Issues, board, PR51/52	Issue/Development/peer paths retained. Owner recorded the final board/label audit in Issue24.
R-10	Labelled Issues14/20/21/22/23/24; C-02/C-06	Gate and failure/review history retained; CI remains mandatory.
R-11	Contributions below; AI disclosure	Both members' linked work/reviews; AI assistance is explicit, not independent human UAT.
R-12	Public repo, dated preflight	Public on13Nov. Team must retain access and service until grading ends.

Actual contribution examples

Member	Engineering / preparation	Peer review / operations
A	SRS/auth/goals; Docker Issue14/PR15; lifecycle work PR35; P0/P1/P2 evidence PR48,PR50,PR53	B bootstrap/gate/release PR26,PR34,PR41,PR51,PR52 reviews; owner protections and supervised deployment/evidence.
B	Feature/Sprint/UI/API contracts, templates and entries PR6,PR7; CI/bootstrap/gate/release work; date guidance PR51; Issue24 index/runbook/PDF	A packaging/digest/P0/preflight/P1/P2 reviews PR15,PR45,PR48,PR49,PR50,PR53; human Cloud Shell bootstrap/readbacks.

Attribution: Codex assisted source/testing/reviews/document assembly. P1 was dispatched by Codex through auto4496 at A's explicit start instruction; student2 credentials were entered directly by the user. These are assisted observations. Both students must read, verify and explain their work; commit counts do not establish participation.

Answer Part 1

CI and Quality Gate

C-01 / R-05: CI workflow handles relevant PRs to main/staging, using an isolated PostgreSQL17 database. PR52 CI43409691975, source b8eb8c9, passed 41 offline tests, 152 API/unit tests in 10 files,38 desktop/mobile browser cases, app build and both Docker targets. Test/migration failure stops later stages; CI success is not live persistence proof.

C-02 / R-06: Active main/staging rules require strict **test-and-docker**, one peer approval, resolved threads and stale-review dismissal. Figure1 shows the native empty bypass/targets; retained gate readbacks establish the lower settings not visible in that crop. Figure2 isolates failed CI after peer approval.

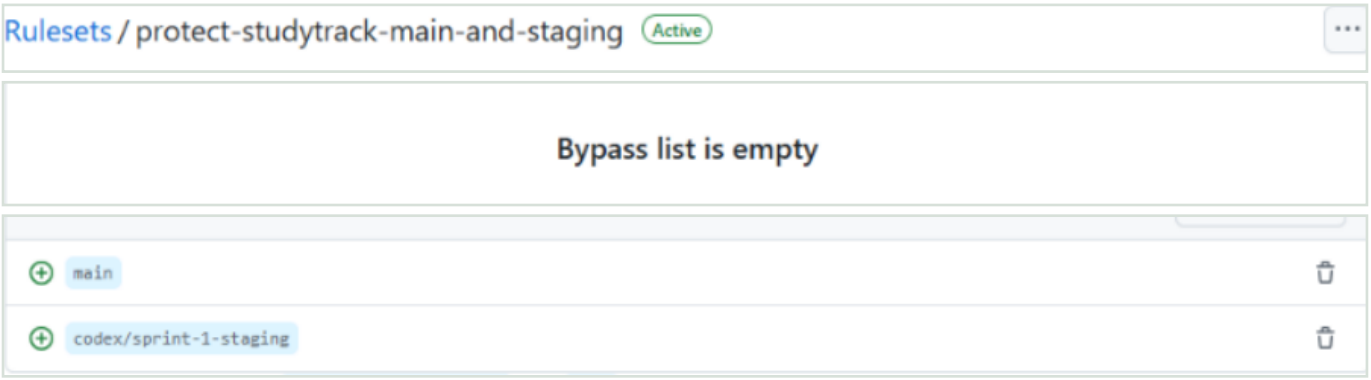


Figure1. C-02 / R-06: owner settings capture shows Active ruleset26925123, empty bypass and main/staging targets. Source: accepted EP6 finalization,12Nov; document panels omit intervening whitespace. Original source. Document crop panels; source unchanged.

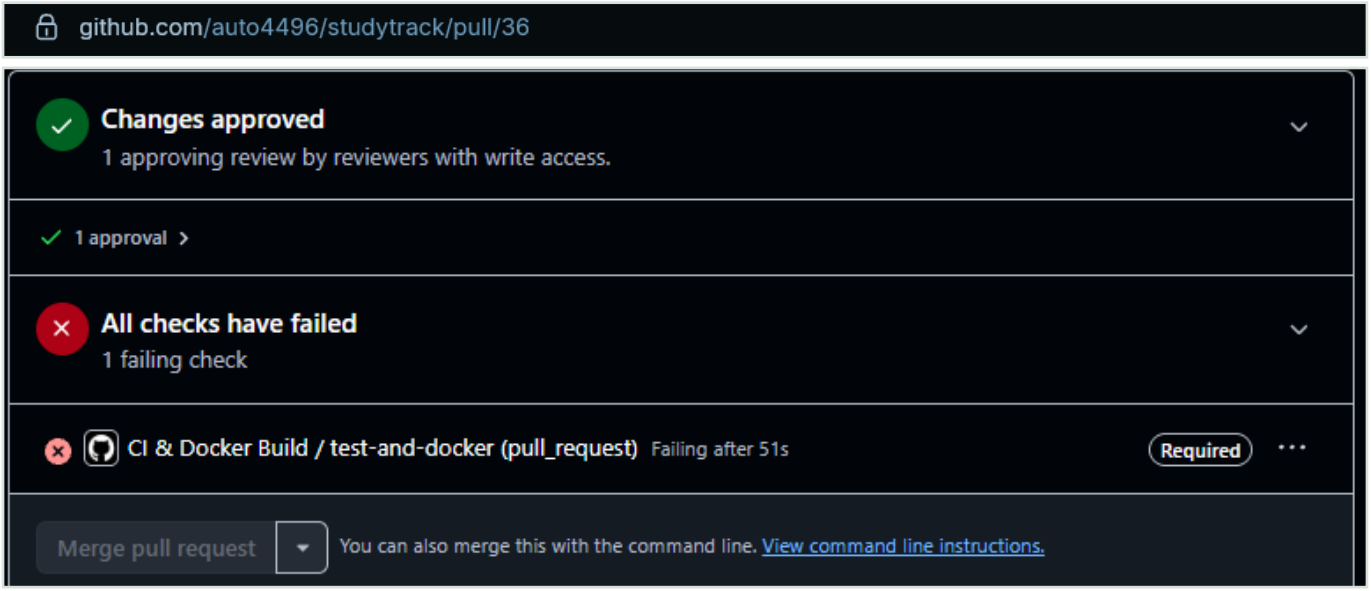


Figure2. C-02 / R-06: native PR36 URL, Changes approved, failed Required test-and-docker and disabled Merge. Exact probe f6330b8; run42781653763. Closed unmerged after capture; no bypass, main change or auto-merge. Original source. Document crop panels; source unchanged.

Probe changed one assertion:45/60 incorrectly expected74, actual75. Hosted result 116 pass / 1 intentional failure; later browser/Docker steps skipped. A approval was limited to the demonstration, explicitly DO NOT MERGE. Cleanup/readbacks preserve history and unchanged main.

Answer Part 2

Docker and Reproducible Configuration

C-03 / R-01,R-02: Dockerfile, .dockerignore, Compose, placeholder .env.example and complete settings/runbook define reproducible configuration. Production runtime serves compiled Express/React as non-root; migration/seed tools are a separate target. Named PostgreSQL volume persists records; no database port is published by the production Compose stack. Local Caddy supplies HTTPS; Cloud Run supplies HTTPS in cloud.

The fresh-clone manifest and no-cache output pin actual tested source **ee56d91dd4abf52f4c6bf13c5080c0f289620cc7**. 35 HTTPS/API/browser/lifecycle checks passed; same local record/session survived app restart, recreation and DB restart. This local45/60=75% record is separate from cloud P0/P1/P2.

Guard correction, source **f938688814f0cadc3caae2a33b58f1ba1a5cf679**, passed 21 guards and35 real production/lifecycle checks with a cached build / existing isolated volume. It was not a new no-cache run. Current Dockerfile additionally labels tools; release CI and Deploy43410699104 built both later targets. Initial runtime-containment failures remain in history; the assertion was preserved while devOptional tooling was excluded.

Representative settings (full table in the runbook)

Setting	Purpose / source	Secret and scope
DATABASE_URL	Prisma; private local config or selected Secret Manager version	Secret; runtime/tools, never report payload
APP_ORIGIN / PORT	Exact HTTPS origin / listen port; service or private local config	Nonsecret; cloud public origin /8080
NODE_ENV / TZ	production / Asia/Bangkok	Nonsecret; workflow/service config
Seed passwords	Generated local config / named secrets	Secret; explicit seed tools only
TEST_DATABASE_URL	Separate disposable test DB with target guard	Secret; tests only, never cloud
CONTAINER_*	Local isolated DB/origin/seed/image source; generated private file	Credentials secret; source/origin metadata nonsecret
GCP_* / WIF_*	demo environment maps project/region/provider/account	Nonsecret references; keyless short-lived token is secret
IMAGE_TAG / VCS_REF	Full reviewed source SHA; workflow/build	Nonsecret provenance; no mutable latest tag

Repeat the isolated local check

```
node scripts/setup-container.mjs
node scripts/container.mjs check --project studytrack-container-review
node scripts/container.mjs up --project studytrack-container-review --fresh-build
node scripts/verify-container.mjs --project studytrack-container-review --browser --lifecycle
node scripts/container.mjs stop --project studytrack-container-review
```

Build/start needs Docker and Node; optional verifier also needs host npm ci and a supported browser. Private generated credentials stay in ignored .env.container.local. Stop retains volumes; no reset/down-with-volumes is a valid persistence proof. Host build/test/migrate/seed commands distinguish local tests, initial seed and normal cloud migration. Report preparation reused the recorded executions without additional local Docker or cloud operations.

Answer Part 3

GCP Deployment and Persistence

C-04 / R-03: Public StudyTrack serves UI/API from Cloud Run **studytrack**, backed by SQL **studytrack-pg** in **studytrack-cpe334 / asia-southeast1**. Database-backed health smoke, A EP5 live journey, P0 save/read and P2 privacy evidence establish actual domain behavior. Browser refresh alone is not C-05 restart/redeploy proof.



Figure3. C-04 / R-03: actual public goal UI confirms saved 20 minutes against 30, 66.67%, stored 12 Nov 2026. Source: P0 on main 3570cd4; original entry saved2026-11-12T14:42:49.190Z, then DTO/refresh checks. This is cloud data, not the local rehearsal. Original source. Document crop panels; source unchanged.

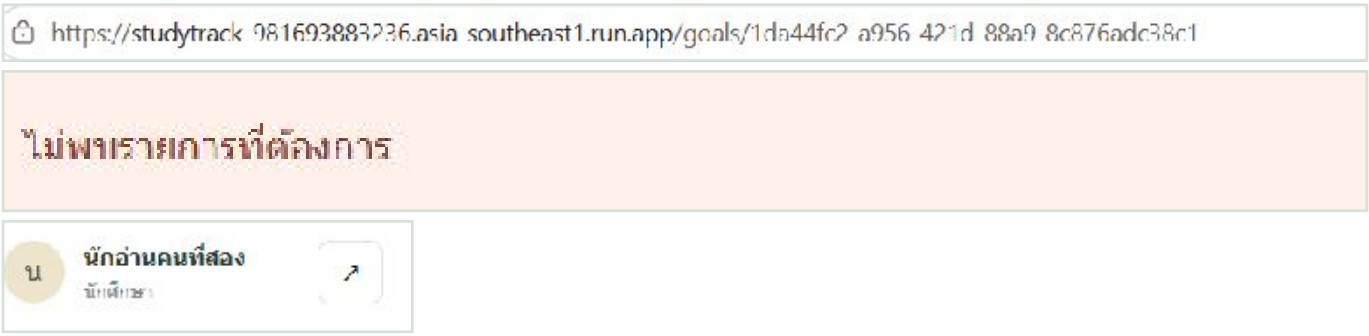


Figure4. C-04 / R-03: same private URL denies authenticated student2 (นักอ่านคนที่สอง shown in the account panel). Source: P2 unobstructed native retake 13 Nov 11:07:20Z, after user-operated sign-in and three API denials; main 6d30aef/run43410699104. Original source. Document crop panels; source unchanged.

Student2 identity and three GET denials show NOT_FOUND and no domain data for goal, entries and fixed-date progress. 404 follows the reviewed contract; raw HTTP status was not independently measured. Reader-one retrieval succeeded before the account switch. No cross-owner write was attempted.

Answer Part 3 - same original record, P0 / P1 / P2

C-05 / R-04: Saved date is **2026-11-12**, not the current browser day. Full P2 comparison equals complete P0/P1 goal, entry-list (including metadata) and progress responses. No baseline record was edited, reseeded, repaired or recreated between phases.



Figure5. C-05 / R-04, P0 BEFORE: original 20/30=66.67%,12Nov2026; main 3570cd4/revision studytrack-g3570cd4a7bca-r43233683592-1. Full public URL/save confirmation is Figure3; raw P0 DTOs retain identity/timestamps. Original source. Document crop panels; source unchanged.



Figure6. C-05 / R-04, P1 AFTER restart-equivalent: original saved-date 20/30=66.67%. Actual same-image run43402554551 created new 100%-serving revision studytrack-g3570cd4a7bca-r43402554551-1; full DTOs match P0. Original source. Document crop panels; source unchanged.



Figure7. C-05 / R-04, P2 AFTER distinct visible redeploy: new selected-date guidance plus original stored-date 20/30=66.67%. Source: main 6d30aef/run43410699104, native UI 13 Nov 10:53:08Z. Selecting 12 Nov changes the read date only. Original source. Document crop panels; source unchanged.

Stable identity: goal **1da44fc2-a956-421d-88a9-8c876adc38c1**; entry **f1146358-2b3a-4aa6-9033-b2d0d95fc542**; count 1; createdAt and updatedAt both **2026-11-12T14:42:49.190Z**; target 30, minutes 20, progress 66.67%,achieved=false. Record values, full responses and native UI agree across all three phases.

Answer Part 3 - operation and database corroboration

P1 actual operation: Run43402554551 reused accepted source 3570cd4 and the same image, skipped build/push/tools/migration/seed, then routed to a new revision. Startup event at 2026-11-13T09:35:54.448330Z is DEPLOYMENT_ROLLOUT with a different instance from preflight. This proves Cloud Run process replacement, not SQL/VM restart or old-instance termination. A explicitly instructed Codex to dispatch through auto4496; no human button click is claimed.

P2 actual operation: Normal run43410699104, source 6d30aef, built both images and waited migration studytrack-migrate-cq47b, with no seed/reset. Selected real helper result verifies lineage/runner/DB binding and smoke. Console images independently corroborate revision/traffic/reference settings.

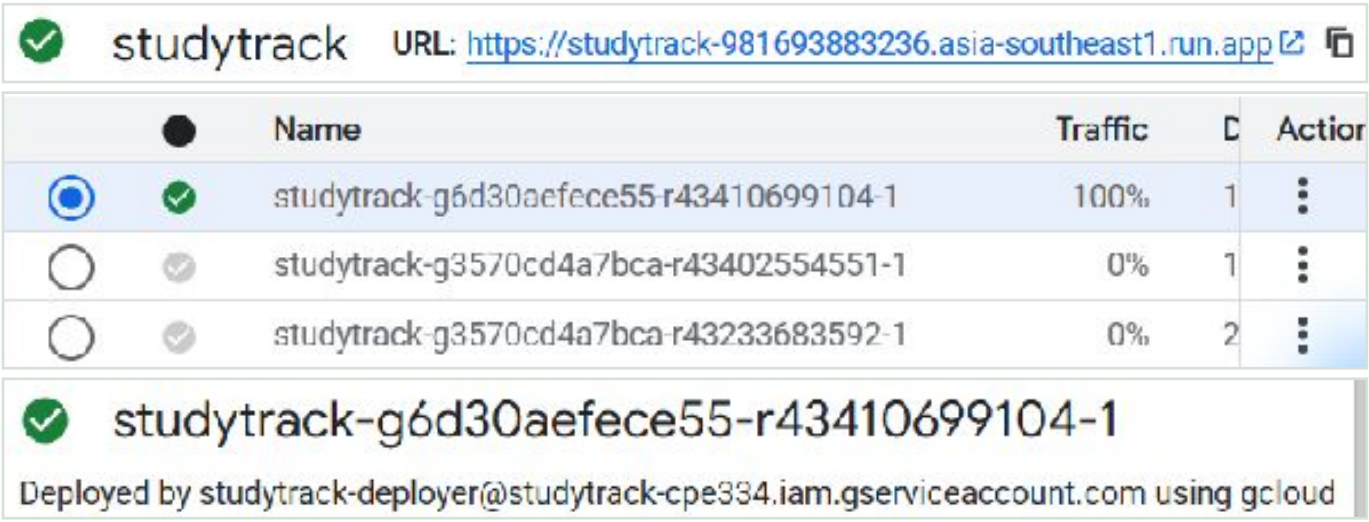


Figure8. C-05 / R-04 and C-06 / R-08: native releaseB service URL, new revision studytrack-g6d30aefece55-r43410699104-1 at 100% and earlier revisions 0%. Source: P2 Console observation,13Nov, actual Deploy43410699104. Original source. Document crop panels; source unchanged.

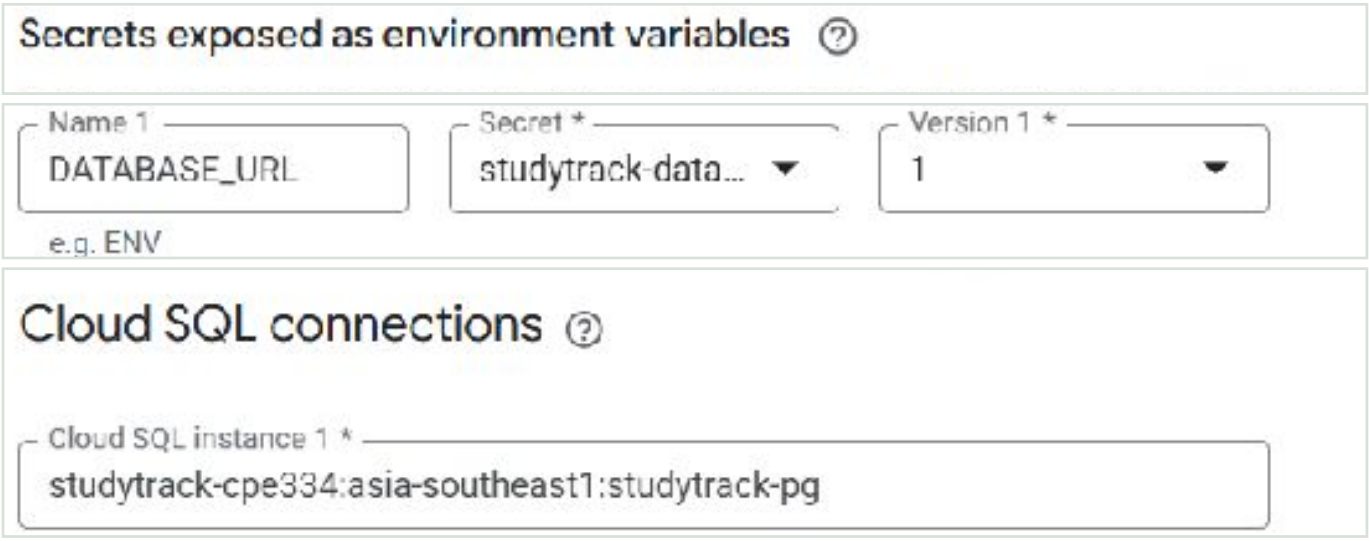


Figure9. C-05 / R-04: native DB secret reference version 1 and SQL attachment studytrack-cpe334:asia-southeast1:studytrack-pg after releaseB. The truncated secret name is corroborated by the exact readback/helper result; no secret value is displayed. Original source. Document crop panels; source unchanged.

P2 record/manifest and B review53 retain original evidence and limits. Database name studytrack is contract-derived; SQL contents/secret payload were not queried for this proof. Optional older-image rollback/roll-forward was not performed. Report content was peer-reviewed and accepted in PR56.

Answer Part 4

Repeatable Release Pipeline

C-06 / R-07 to R-10: One concrete labelled change was B's selected-date reading guidance under Issue22. Specification preceded the change; it adds visible help and an accessible description, without changing domain data/API behavior.

Stage	Direct evidence / source	Actual acceptance / result
Feature -> staging	PR51, head 99d4908; CI43408783283	A specific review/approval; CI PASS; staging b8eb8c9. Local fixture figures were not cloud proof.
Separate release -> main	PR52, fresh CI43409691975	A separately approved release; main 6d30aef tree59ca1b1113958c7f6fc14d999fb4ea8ed84cffab equals accepted tree.
Main -> pipeline	Deploy43410699104, attempt1,push	Normal CI/build runtime+tools/push/WIF/migration/ready/traffic/smoke SUCCESS.
Pipeline -> public	Release result, Figures7-9	New visible guidance; studytrack-g6d30aefece55-r43410699104-1 at 100%; same original saved data.

Runtime index **54dbdf5fb0fa4e6f7baec37f0568bbcff4db0fd97a0550df8c24f9923e80ba62**; resolved platform **e311c6c11c71d51921464217ec498eef2708ccafa1d4d6f346aaec2b40a15cca** (both sha256). Service index and revision platform are different levels of the verified same lineage. Earlier failed run43197450278 remains FAILED; PR45 correction and successful releaseA run43233683592 retain the diagnosis rather than suppressing the guard.

Repeatable procedure and lookup

Deployment runbook: labelled linked Issue -> feature/fix -> peer-approved protected staging -> separate peer-approved main release -> main-triggered pipeline. deploy.yml and release.py validate clean/current main, image source/lineage and existing config, wait migration without seed, create no-traffic revision, check Ready/digest/config, then route and smoke. Normal updates do not rerun initial IAM/SQL/secret bootstrap.

Record actual run headSha, registry source/index/platform and serving traffic/revision; safe lookup commands avoid full env/secret dumps. Initial bootstrap was supervised in Cloud Shell; later WIF/demo uses a short-lived token with no stored service-account key. Never bypass failed CI/guards or self-approve a release.

Availability and submission

The report/index/runbook were accepted through PR56 and integrated into protected staging311452c. At 13 Nov 2026, 13:31Z, preflight found repo public, main6d30aef unchanged, strict gate active and public health/UI smoke PASS. The owner's later final-package check records public availability, protections and board/label verification. These are dated observations; keep the repo and service available until grading ends.

Submission filename: **lab5_secB_TH-057_P8.pdf**. Handout deadline: **15 November 2026, 23:59 Bangkok**; confirm the current LEB2 activity before upload. The actual upload receipt is tracked separately in Issue24. Native source images remain unchanged; cropped panels are labelled, secret payloads are absent, and AI-assisted execution/limitations are disclosed.